

PRACTICAL EXAM

Course: AI Algorithms and Predictions

Dataset: "Moteur.ma_SDBDIA2A"

Prof: Abdellah TAHIRI, Ph.D

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Problem Statement:

We aim to develop a data-driven tool that helps users to estimate the fair price of a second-hand car based on: historical data, car features, and market trends.

- ◆ **Data Sources:** <https://www.moteur.ma>
- ◆ **Features to collect:** Brand / Model; Year; Mileage; Fuel type; Transmission; Engine power; Location; Condition; Price (target variable);
- ◆ **Key Value:** Helps buyers to avoid overpaying and helps sellers to price cars competitively.

The approach should follow the plan outlined below:

PART A – Data Collection

- Extract useful data from the website:
<https://www.moteur.ma/fr/voiture/achat-voiture-occasion>
- Libraries for web Scraping: requests; BeautifulSoup; Selenium; ...

PART B – Data Exploratory Analysis

- Understand data.
- Visualize Data.
- Extract insights.
- Prepare for modeling.

PART C – Data cleaning and preprocessing

- Handle Missing Values.
- Handle Outliers.
- Encode Categorical Variable.
- Features selection and engineering.
- Data transformation: normalization and standardization.

PART D – Models Building

- Define Features & Target.
- Train-Test Split.
- Define and fit models.
- Make Predictions.

PART E – Model Evaluation and validation

- Evaluate model.
- Compare models performance.
- Tune Hyperparameters.
- Cross validation.
- Model selection.



PART F – Interpretation and communication of results

- Provide different business insights based on your analysis that may help user for better decision-making.